

# EXAM PAPER: CS101 - Introduction to Programming

Total Duration: 45 Minutes | Total Marks: 20 Marks

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**Q1. [Marks: 2 | Time: 90s] What is the output of `print(2 ** 3)` in Python?**

- A) 5
- B) 6
- C) 8
- D) 9

**Correct Answer: C**

**Q2. [Marks: 2 | Time: 90s] Which data structure follows First-In First-Out (FIFO)?**

- A) Stack
- B) Queue
- C) Tree
- D) Graph

**Correct Answer: B**

**Q3. [Marks: 6 | Time: 300s]**

Question: Explain the difference between a list and a tuple in Python.

**Key Points:**

- Lists are mutable whereas tuples are immutable.
- Lists use square brackets `[]` whereas tuples use parentheses `()`.
- Tuples are faster and consume less memory than lists.

**Q4. [Marks: 10 | Time: 600s]**

Question: Write a Python program logic to check if a string is a palindrome.

**Key Points:**

- Convert string to lowercase and remove spaces/special characters.
- Compare the string with its reverse (`s == s[::-1]`).
- Return True if identical, otherwise return False.